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Integrative biomechanics of a human–robot carrying task: implications for future collaborative work

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Abstract
Patients with sarcopenia, who face difficulties in carrying heavy loads, may benefit from collaborative robotic assistance that is modeled after human–human interaction. The objective of this study is to describe the kinematics and spatio-temporal parameters during a collaborative carrying task involving both human and robotic partners. Fourteen subjects carried a table while moving forward with a human and a robotic partner. The movements were recorded using a three-dimensional motion capture system. The subjects successfully completed the task of carrying the table with the robot. No significant differences were found in the shoulder and elbow flexion/extension angles. In human–human dyads, the center of mass naturally oscillated vertically with an amplitude of approximately 2 cm. The here presented results of the human–human interaction serve as a model for the development of future robotic systems, designed for collaborative manipulation.

Keywords Human–human interaction · Assistive robot · Human–robot cooperation · Kinematics · Spatio-temporal parameters · Joint handling

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1 Introduction

Reduced independence and quality of life are impacts on elderly individuals, caused by symptoms of sarcopenia, a progressing skeletal muscle disease (Cruz-Jentoft et al., 2018). The quality of life and independence of elderly participants were increased in previous studies through adjustment of the home environment and implementing strategies to reach the participants' self-defined individual goals (Szanton et al., 2021). Robots for object manipulation or cleaning could be an additional possibility to increase the quality of life and independence of the elderly in the future, since it is assumed that robots will enter the domestic environment (Fortunati et al., 2015; Haddadin & Croft, 2016). Since sarcopenia patients often report difficulties in carrying heavier loads (Janssen et al., 2002), robots could assist through collaborative manipulation of heavy objects. As there are manifold possibilities for collaborative manipulation, it can be determined by contact points between body and object and orientation of the dyads (Agravante et al., 2019), such as in carrying a table in front of the body, facing front to each other, and moving forward. Robots constructed for collaborative manipulation face the challenge of integrating a human's unpredictable behavior into the control loop (Agravante et al., 2013). To develop such a controller, human–human interaction serves as a model, whereby humans have multiple ways of interacting, such as visual, verbal and haptic communication. Experimental studies on human dyads restricted to haptic communication demonstrated that carrying a large extended object collaboratively was successful, although the velocity was slower compared to unrestricted communication (Jensen et al., 2021). Several studies have already shown that collaborative manipulation with a robotic partner can be performed successfully communicating via haptic-only (Bussy et al. 2012a, 2012b; Ćehajić et al. 2017; Mielke et al. 2020; Mörtl et al. 2012), haptic and vision (Agravante et al., 2014; Kosuge et al., 2000; Sirintuna et al., 2022; Yokoyama et al., 2003), and vision-only (Mielke et al., 2020; Stückler & Behnke, 2011). A vision-based controller for human–robot collaborative manipulation, reacting immediately to the human's input by motion-prediction based on the kinematics of the collaboratively carried table, turned out to be significantly slower than human dyads without restrictions (Mielke et al., 2020). Thus, the effects of reduced velocity, leading to changes in kinematics and spatio-temporal parameters such as reduced step length (Fukuchi et al., 2019), should be considered. Since step length seems to be influenced by slower movement, the effects of decreased step length on kinematics, such as changes of the range of motion (ROM) of the lower limb's joints for shorter steps, should be considered (Lim et al., 2017). To date, no information is available about the kinematics and spatio-temporal parameters of human dyads carrying a table in front of the body moving forward, which is important as a model for human–robot interaction. Further, the effects of the robot on the human's kinematic and spatio-temporal parameters are unknown.

Therefore, this study aims to describe the kinematics and spatio-temporal parameters of a human dyad carrying the table in front of the body with the leading human moving forward. This data can be used as a model for human–robot interaction. Additionally, the kinematics and spatio-temporal parameters of a human–robot interaction are described, in which the human partner carried a table in front of the body with a robotic partner.

It is hypothesized that there are no differences in upper body kinematics when interacting with a robotic partner compared to human–human interaction. However, for the spatio-temporal parameters and lower body joint kinematics, we anticipate differences due to a slower walking velocity. Specifically, it is hypothesized that step frequency, step length and velocity are smaller for the interaction with the robotic partner compared to human–human interaction, whereas task duration, ground contact time, and duty factor are larger for the interaction with the robotic partner compared to human–human interaction. For the hip and knee joints, a smaller ROM is expected whereas the ROM for the ankle joint is expected to be larger for the interaction with the robotic partner.

2 Methodology

2.1 Subjects

14 volunteers (30.2 ± 9.9 years, 174.6 ± 9.7cm body height, and 68.8 ± 13.1 kg body mass) were included in the study. The genders were evenly distributed (seven male and seven female). Informed written consent to participate in this study was provided by all subjects, and the study was approved by the ethics committee of the German Sport University, Cologne (No 132/2020). The experiments were performed at KUKA Deutschland GmbH, Augsburg, Germany.

2.2 Study design and experimental protocol

The task of the experiment was to carry a table in front of the body while moving forward, first with a human partner then with the robotic partner. The experimental area was set up as if a table (L × W × H = 120 × 75 × 73 cm, 14.35 kg) was carried from one room to another through an opening of 190 cm (Fig. 1). The start and end of the task were marked on the floor. Communication was not restricted in the human-only trial, while the interaction with the robot was primarily based on haptic information. The assistive robot consisted of a three degrees of freedom KUKA mobile platform and two six degrees of freedom KUKA LBRiiwa robotic arms,

which were mounted on the mobile platform (Bender et al., 2021). The robot was guided by torques and forces (Braun et al., 2021; Ritter & Sharma, 2019). The table is gripped by the robot using a system consisting out of a passive gripper and a torque limiter. This combination enables the transfer of torques but ensures at the same time, that the system is not damaged if the torque is too high. Additionally, the robot is equipped with a 3D-sensor, as well as laser and radar enabling the robot to recognize its environment; thus, the robot knows its position in the room and detects obstacles and objects. Safety strategies are implemented which cause the robot to reduce its velocity if the system, consisting of robot, human and table, gets too close to an obstacle or object. A detailed description of the robot can be found elsewhere (Braun et al., 2021). The interaction between robot and human is described as collaborative, since both work together to accomplish the task of carrying the table through a joint team effort (McGirr et al., 2022). Six repetitions were recorded for each scenario. The first three repetitions served to get familiar with the task. From the remaining three repetitions, only the trial with the shortest time interval between lifting and setting down the table was evaluated, since it is assumed to be the trial with the best performance. Additionally, to investigate the trust in robots during the human–robot interaction, a questionnaire was given to the subjects. These results are published elsewhere (Bender et al., 2021).

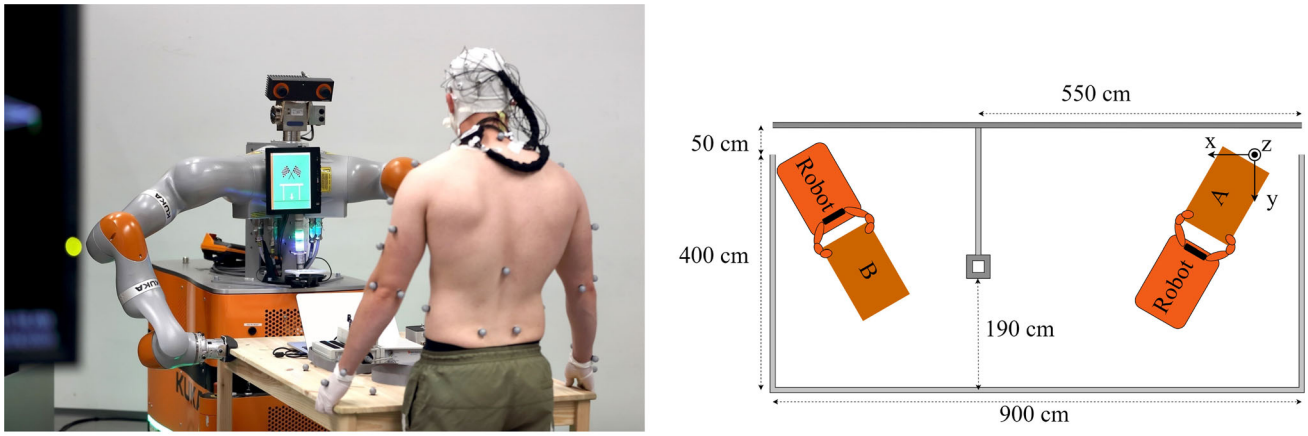
2.3 Data collection

The subject's movements were recorded by a three-dimensional motion capture system (Vicon Motion Systems Ltd, Oxford, UK) with an acquisition frequency of 100 Hz. Therefore, 15 infrared MX13 cameras were used, to cover the complete measurement volume of the experimental area. To capture the motion of the subject, 56 retro-reflective spherical markers (diameter: 19 mm) were positioned on the subject's body. The markers were attached to upper and lower body bony landmarks, and technical clusters, consisting of three markers, were added laterally on the thigh, shank, and upper arm segments. The supplementary material provides a list with all markers used (Table S.I). Prior to data collection, a static reference was recorded with the subject standing upright. Zero degrees of the ankle, knee, and hip joints were defined by the static reference position; the shoulder joints were 90° abducted.

2.4 Data processing

The 3D marker trajectories were labeled automatically with a predefined virtual model (Vicon Nexus 2.11, Vicon Motion Systems, Oxford, UK). The rigid body fill function was used to fill the gaps. If not enough markers of the corresponding segment were visible, the pattern fill function was chosen to close the remaining gaps. The marker trajectories were filtered with a fourth-order, zero-lag low pass Butterworth filter with a cut-off frequency of 10 Hz. Touch Down (TD) and Foot Off (FO) gait events were set manually by a trained researcher through visual inspection of the 3D marker trajectories. TD was defined by the first frame, in which either the marker of the toe or the heel stopped moving in the vertical and horizontal direction after the swing phase. The FO event was defined by the first frame where the marker of the heel and the toe started moving in the vertical or horizontal direction continuously after the stance phase. Previously, Ghoussayni et al. (2004) demonstrated that these criteria are reliable.

Task duration described the elapsed time between first FO and the last TD. Step frequency was determined by the number of detected steps divided by the task duration, whereas the mean step length was determined by the number of detected steps divided by the distance covered by the center of mass (COM), which was calculated as the incremental sum of horizontal distances. The step width was calculated as the lateral distance between both heel markers with respect to the pelvic local coordinate system, as described below. The ground contact time was determined by the elapsed time between TD and FO events of the right foot. Duty factor was determined by the ratio of the ground contact times of the left and right feet within one gait cycle and the time of the total gait cycle (Fihl & Moeslund, 2007). The overall subject's COM position was determined based on Minetti et al. (1993) with the segments defined by Dempster et al. (1955). Velocity was calculated by the time incremental derivation of the position of the COM. The vertical displacements of the COM and hand were calculated as the differences in height to initial ground contact in each gait cycle. The COM–hand distance was calculated as the distance between the COM and the midpoint between the right and left hand, to describe the distance between human and table. The local coordinate systems for the thorax, upper arm, and forearm were determined according to the recommendations of the International Society of Biomechanics (Wu et al., 2005), and the local coordinate systems for the pelvis, thigh, shank, and foot were set according to Zatsiorsky (1998) and Hamill et al. (2014). The joint centers of the hip joints were set according to Harrington et al. (2007), and the joint centers of the glenohumeral joints were set according to Rab et al. (2002). Shoulder flexion/extension, abduction/adduction, and internal/external rotation and elbow flexion/extension were calculated according to the recommendations of the International Society of Biomechanics (Wu et al., 2005). The joint angles of the lower extremity (ankle dorsi–/plantarflexion, knee flexion/extension, hip flexion/extension) were calculated according to Zatsiorsky (1998) and Hamill et al. (2014). Additionally, thorax flexion/extension and rotation were calculated between the pelvic and thorax local coordinate systems.



(a) (b)

Fig. 1 Experimental setup. **a** Assistive robot and subject carrying the table collaboratively (Copyright: 2020 KUKA Deutschland GmbH, Augsburg, Germany). **b** Top view of the experimental area. Positions A and B show the two positions of the robot and table to start and end the task, respectively. The light grey area indicates mobile partitions, which limit the test area. The rectangle in the middle indicates a pillar and the dark grey line the laboratory wall. The origin of the global coordinate system is positioned on the floor at table position A

COM and hand vertical displacement and COM–hand distance were determined for each gait cycle, normalized to time and averaged per subject in each condition. COM and hand amplitudes were calculated for each gait cycle by finding the difference between the maximum and minimum values. The resulting values were averaged per subject for each condition. COM and hand height were determined by the average height in each gait cycle and averaged for each subject in each condition. Ground contact time, duty factor, and step width were determined for each gait cycle. Afterwards, their respective averages were calculated. For each gait cycle and stance phase, the time-normalized joint angle time courses for the upper and lower body were calculated and averaged per subject and per condition. The respective ROM was calculated by finding the difference between the maximum and minimum joint angle values for each gait cycle or stance phase. These values were then averaged per subject and per condition. For each gait cycle or stance phase, the average joint angles of both the upper and lower body were calculated. These values were then averaged per subject and per condition. To reduce the amount of data, only the right side was analyzed for the upper and lower body joint angles and spatio-temporal parameters.

2.5 Statistical analysis

Statistical analysis was performed using R Studio (version 2023.03.1) with R version 4.1.2. The parameters were tested for normal distribution using the Shapiro test with the level of statistical significance set to $\alpha = 0.05$. Accordingly, a two-tailed paired t-test or a two-tailed paired Wilcoxon test for not normally distributed parameters were performed, to check the conditions for differences with the significance level set to $\alpha = 0.05$. The data is reported as the mean (standard deviation) and median (interquartile range) for each condition (Sainani, 2012). Cohen's d effect size, defined as small (0.2), moderate (0.5), and large (0.8) (Cohen, 1988), was calculated for the normally distributed data by using the difference scores (King et al., 2018). An equivalent to effect size was calculated for not normally distributed data by the matched pairs rank-biserial correlation (King et al., 2018). To identify differences between the conditions in parameters with time courses, a Statistical Parameter Mapping (SPM) analysis with a two tailed t-test with the level of significance set to $\alpha = 0.05$ was performed. The spm1d toolbox (version M.0.4.10) for Matlab (version 2022b) was used (Pataky, 2010).

3 Results

3.1 Spatio-temporal parameters

Fulfilling the task of carrying the table with the assistive robot took significantly longer ($P < 0.001$) and the velocity was significantly slower ($P < 0.001$) compared to the interaction with the human partner (Table 1). Duty factor ($P < 0.001$) and ground contact time ($P < 0.001$) were significantly higher with the assistive robot compared to human–human interaction (Table 1). Step length ($P < 0.001$), step frequency ($P < 0.001$), and step width ($P = 0.01$) (Table 1) were significantly smaller for the interaction with the robot compared to human–human interaction.

Table 1 Spatio-temporal parameters while carrying a table forward in front of the body with another human or an assistive robot and the differences between both conditions

Parameter	Human M (SD)	Robot M (SD)	Diff. M (SD)	95% CI	Test statistics	*P*	Cohen's d_z
Velocity (m/s)	0.69 (0.14)	0.26 (0.09)	−0.44 (0.14)	0.4 to 0.5	t(13) = 12.05	<0.001	d = 3.22
Task duration (s)	10 (2)[a]	35 (18)[a]	24 (16)[a]	−39.9 to −18.6[b]	W = 105(+105,−0)	<0.001	r = −1
Step width (mm)	205 (42)	160 (39)	−45 (56)	13.0 to 77.1	t(13) = 3.03	0.01	d = 0.81
Step length (mm)	491 (111)	295 (64)	−195 (114)	129.4 to 261.0	t(13) = 6.41	<0.001	d = 1.71
Step frequency (1/min)	91 (14)	55 (17)	−35 (17)	25.3 to 45.3	t(13) = 7.64	<0.001	d = 2.04
Ground contact time (s)	0.82 (0.19)[a]	1.64 (1.07)[a]	0.75 (0.83)[a]	−1.3 to −0.5[b]	W = 105(+105,−0)	<0.001	r = −1
Duty factor	0.61 (0.03)	0.77 (0.06)	0.15 (0.04)	−0.2 to −0.1	t(13) = −12.74	<0.001	d = 3.4
COM height (mm)	1032 (103)[a]	1032 (90)[a]	−2 (17)[a]	−0.6 to 12.4[b]	W = 105(+31,−74)	0.194	r = 0.41
COM amplitude (mm)	23 (6)	17 (5)	−6 (6)	2.3 to 9.4	t(13) = 3.53	0.004	d = 0.94
Hand height (mm)	916 (59)	886 (24)	−29 (47)	2.1 to 56.9	t(13) = 2.32	0.037	d = 0.62
Hand amplitude (mm)	22 (7)[a]	15 (4)[a]	−6 (6)[a]	3.3 to 16.1[b]	W = 105(+3,−102)	<0.001	r = 0.94
COM–hand distance (mm)	175 (32)	158 (21)	−17 (34)	−3.0 to 36.6	t(13) = 1.83	0.09	d = 0.49

M mean, *SD* standard deviation, *Diff.* mean/median deviation of robot to human carrying partner, *CI* confidence interval, *P P* value, *W* sum of signed ranks (sum of positive, sum of negative ranks), *t* t-value, *d* Cohen's d effect size, *r* matched pairs rank-biserial correlation, *COM* center of mass. [a] Wilcoxon singed rank test applied, and values are expressed as median (interquartile range); [b] Non-parametric confidence interval

The amplitude of the COM was significantly smaller ($P = 0.004$) with the assistive robot compared to human–human interaction (Table 1). No differences between the two conditions were found for the height of the COM (Table 1). The SPM analysis of the COM vertical displacement showed a significantly larger COM vertical displacement for the human–human interaction from 22 to 43% ($P < 0.001$) and from 72 and 91 % ($P < 0.001$) of the gait cycle (Fig. 2b). The SPM analysis showed a significantly larger hand vertical displacement between 4 and 13 % ($P = 0.009$) and from 50 and 66 % ($P < 0.001$) of the gait cycle for the human–robot interaction, whereas for the interval between 28 and 39% ($P = 0.004$) of the gait cycle the hand vertical displacement was significantly larger for the human–human interaction (Fig. 2d). The average hand height and amplitude were significantly smaller ($P = 0.037$, $P < 0.001$, respectively) for the interaction with the robotic partner (Table 1). The average COM–hand distance and the corresponding SPM analysis did not show any significant differences between the two conditions (Fig. 2f, Table 1).

3.2 Joint kinematics of the upper body

A significantly larger thorax flexion for the human–robot interaction was found for whole gait cycle ($P = 0.004$) (Fig. 3e). The average joint angle of the thorax flexion was significantly larger for the interaction with the robot, compared to the human–human interaction ($P = 0.005$) (Table 2). The ROM did not show any significant differences between the two conditions (Table 2). The SPM analysis and the average angle of the thorax rotation showed no significant differences between the two conditions. The ROM of the thorax rotation was significantly larger ($P = 0.003$) for the interaction with the robotic partner compared to the human–human interaction. The SPM analysis, the average joint angle, and the ROM showed no statistically significant differences for the shoulder flexion between the two conditions (Fig. 3a, Table 2). A significant larger shoulder abduction was found in the SPM analysis for the human–robot interaction for the complete gait cycle ($P < 0.001$) (Fig. 3b). The average joint angle and the ROM for shoulder abduction were significantly larger for the interaction with the robotic partner ($P = 0.002$, $P < 0.001$, respectively) (Table 2). A significant larger shoulder internal rotation was found for the human–robot interaction in the range of 54 to 100% ($P = 0.02$) of the gait cycle (Fig. 3c). The average joint angle of the shoulder rotation was significantly larger for the robotic partner ($P = 0.014$), and the ROM did not differ significantly (Table 2). The SPM analysis of the elbow flexion (Fig. 3d) did not show any significant differences, nor did the average elbow joint angle (Table 2). The ROM of the elbow joint was significantly larger ($P = 0.019$) for the human–robot interaction compared to the human–human interaction (Table 2).

3.3 Joint kinematics of the lower body

The SPM analysis showed a significant larger dorsalextension for the human–robot interaction for the ankle joint in the range of 12 to 47% ($P = 0.002$) of the stance phase, whereas in the late stance phase (from 74 to 100%) the dorsalextension of the ankle joint during the human–robot

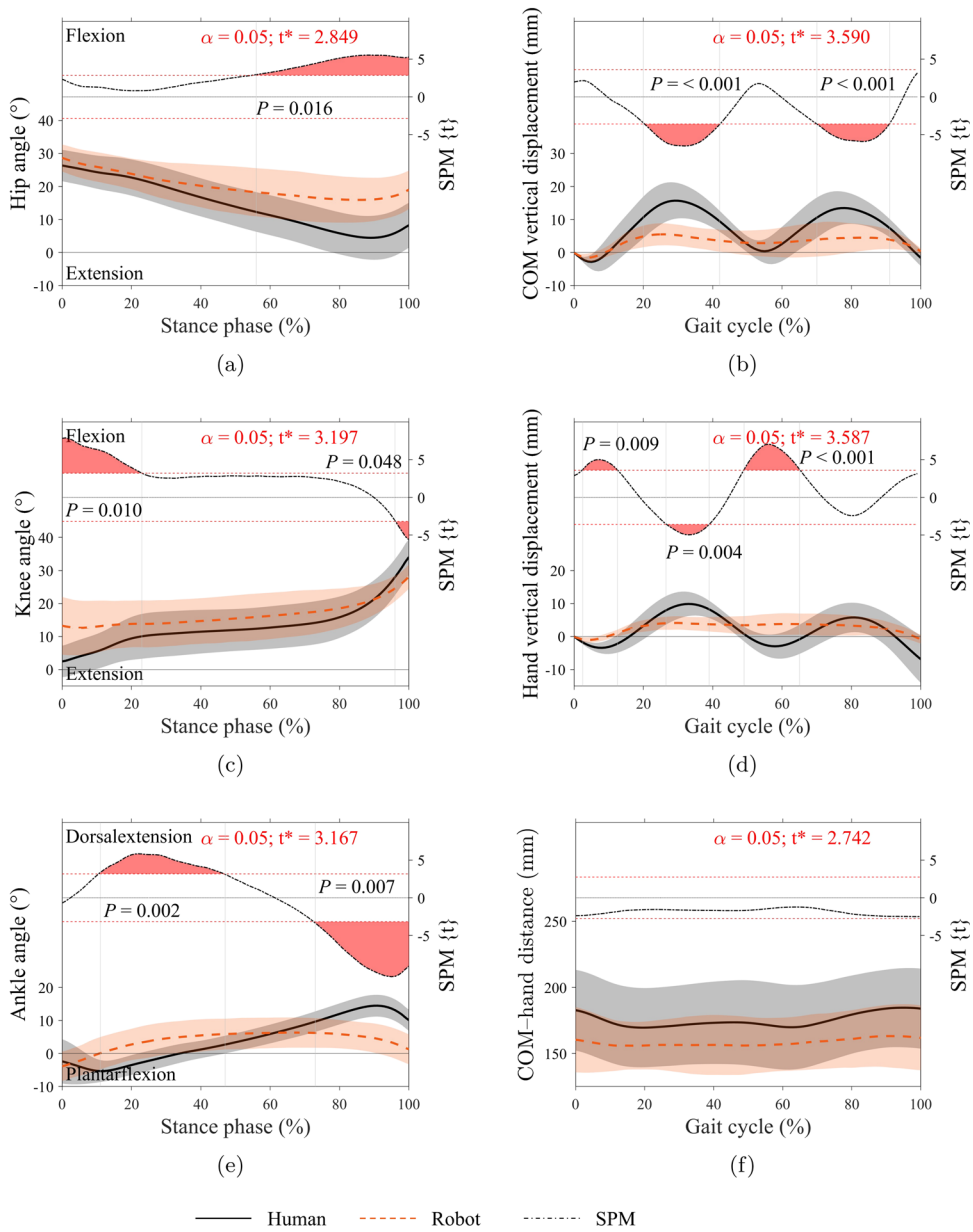


Fig. 2 SPM analysis of lower body and COM. Figures show the mean values of the hip joint flexion/extension (**a**), knee joint flexion/extension (**c**), and ankle joint dorsalextension/plantarflexion (**e**) as a function of normalized stance phase (left y-axis) and the mean values of the COM vertical displacement (**b**), hand vertical displacement (**d**), and COM–hand distance (**f**) as a function of normalized gait cycle (left y-axis) with colored area representing the standard deviation of carrying a table with another human (black solid line) or an assistive robot (orange dashed line). In the upper part, the corresponding SPM inference (black dotted line) with the ranges for significantly larger or smaller values colored red is shown (right y-axis). The statistical significance level (α), critical thresholds (t*) and, in case of significant difference(s), the cluster-level probability value(s) (P) are displayed

interaction was significantly smaller ($P = 0.007$) (Fig. 2e). The average ankle joint flexion did not differ significantly and the ROM was significantly smaller ($P = 0.015$) for the interaction with the robot compared to human–human interaction (Table 2). For the knee joint, a significant larger flexion for the human–robot interaction was observed from 0 to 24% ($P = 0.01$) of the stance phase, and a significant smaller flexion for the human–robot interaction was observed from 98 to 100% ($P = 0.048$) of the stance phase (Fig. 2c). The average knee joint angle was significantly larger ($P = 0.005$) for the interaction with the robot compared to the human–human interaction, and the ROM was significantly smaller ($P < 0.001$) for the interaction with the robot compared to the human–human interaction (Table 2). A significantly larger hip flexion was observed for the human–robot interaction from 56 to 100% ($P = 0.016$) of the stance phase (Fig. 2a). The average hip joint flexion was significantly larger ($P = 0.006$) for the interaction with the robot compared to the human–human interaction. The ROM was significantly smaller ($P < 0.001$) for the interaction with the robot compared to the human–human interaction (Table 2).

4 Discussion

The present study aims to describe the upper and lower body kinematics and spatio-temporal parameters of a human dyad

Table 2 Postural parameters while carrying a table forward in front of the body with another human or an assistive robot and the differences between both conditions.

		Human M (SD)	Robot M (SD)	Diff. M (SD)	95% CI	Test statistics	*P*	Cohen's d_z
Ankle joint dorsiflexion	Average	4 (3)	4 (4)	0 (4)	−2 to 2.1	t(13) = 0.02	0.987	d = 0
	ROM	22 (5)	17 (4)	−5 (6)	1.1 to 8.5	t(13) = 2.8	0.015	d = 0.75
Knee joint flexion	Average	13 (6)	16 (7)	3 (4)	−5.6 to −1.2	t(13) = −3.37	0.005	d = 0.9
	ROM	32 (4)	21 (6)	−11 (5)	8.2 to 13.4	t(13) = 8.87	<0.001	d = 2.37
Hip joint flexion	Average	14 (6)	20 (6)	6 (6)	−9.3 to −2	t(13) = −3.3	0.006	d = 0.88
	ROM	23 (4)	16 (3)	−7 (5)	3.8 to 10	t(13) = 4.84	<0.001	d = 1.29
Thorax flexion	Average	2 (8)[a]	7 (10)[a]	2 (9)[a]	−7.6 to −0.8[b]	W = 105(+95,−10)	0.005	r = −0.81
	ROM	4 (1)	5 (1)	1 (2)	−2.5 to 0.2	t(13) = −1.89	0.081	d = 0.51
Thorax left rotation	Average	3 (3)	2 (4)	−2 (4)	−0.4 to 3.8	t(13) = 1.73	0.107	d = 0.46
	ROM	8 (3)[a]	12 (3)[a]	3 (4)[a]	−4.6 to −1.4[b]	W = 105(+97,−8)	0.003	r = −0.85
Shoulder joint flexion	Average	−7 (11)	−5 (8)	2 (13)	−9 to 5.9	t(13) = −0.46	0.656	d = 0.12
	ROM	9 (3)	11 (5)	2 (4)	−4.5 to 0.1	t(13) = −2.09	0.057	d = 0.56
Shoulder joint abduction	Average	15 (5)	21 (5)	6 (5)	−8.6 to −2.5	t(13) = −3.9	0.002	d = 1.04
	ROM	7 (2)	10 (4)	3 (3)	−4.5 to −1.6	t(13) = −4.56	<0.001	d = 1.22
Shoulder joint internal rotation	Average	8 (12)	16 (13)	8 (10)	−13.2 to −1.8	t(13) = −2.84	0.014	d = 0.76
	ROM	9 (3)	9 (3)	0 (5)	−3.2 to 2.2	t(13) = −0.39	0.699	d = 0.11
Elbow joint flexion	Average	50 (18)	50 (11)	0 (22)	−12.6 to 12.4	t(13) = −0.02	0.984	d = 0.01
	ROM	8 (4)	12 (4)	4 (6)	−7.1 to −0.8	t(13) = −2.67	0.019	d = 0.71

Mean and standard deviation values/medians and interquartile ranges are given in degrees. The mentioned angle direction is the positive angle. The negative value is the angle for the opposing direction. *M* mean, *SD* standard deviation, *Diff.* mean/median deviation of robot to human carrying partner, *CI* confidence interval, *P P* value, *W* sum of signed ranks (sum of positive, sum of negative ranks), *t* t-value, *d* Cohen's d effect size, *r* matched pairs rank-biserial correlation, *ROM* range of motion. [a] Wilcoxon signed rank test applied, and values are expressed as median (interquartile range); [b] Non-parametric confidence interval

carrying a table collaboratively in front of the body moving forward to serve as a model for human–robot collaboration. Furthermore, all participants successfully completed the task of carrying the table from one room to another through an opening of 190 cm in collaboration with the robot, with the leading human moving forward.

4.1 Human–human collaborative manipulation

The results of the present study indicate that carrying a table collaboratively with a human partner in front of the body moving forward seems to be associated with slower velocities compared to the reported preferred velocities for normal unconstrained walking (Table SII) (Öberg et al., 1993; Riley et al., 2001; Grasso et al., 1998). Sghaier et al. (2022) investigated the collaborative manipulation of an object carried with only one hand sideways the body with both subjects moving forward and reported a walking velocity of 1.4 ± 0.14 m/s, which is about twice as fast as the velocity in the present study. Carrying a table collaboratively in front of the body means that the other partner has to walk backwards, which might have limited their walking velocity, since walking backward is known to result in slower velocities compared to walking forward (Laufer, 2005). According to the slow velocity (0.69 m/s) in the present study, the observed step length is relatively short (49.1 cm) but is in line with the reported step lengths of 51.5 cm for normal unconstrained walking at a slightly faster velocity (0.87 m/s) (Table SII) (Öberg et al., 1993). Velocity, step length, and step frequency are mutually dependent; thus, the step frequency must be smaller compared to normal unconstrained walking, which is confirmed by our results (Table 1, Table SII). Ground contact time usually increases with decreasing velocity (Lewek, 2011; Yang et al., 2013). Compared to other studies, the ground contact time observed in the present study seems to be rather short for the slow velocity (Table SII). A previous study showed that the duty factor reduces for higher velocities and ranges between 0.58 and 0.7 for walking (Fihl & Moeslund, 2007). The duty factor observed in the present study is associated with higher walking velocities, despite a slower velocity. The differences found between the reported results of unconstrained walking and the results of carrying a table collaboratively with another human of the present study, also seem to affect the natural oscillation of the COM since the amplitude of the oscillation found in the present study (23(6) mm) is outside the range of previously reported COM vertical displacement of

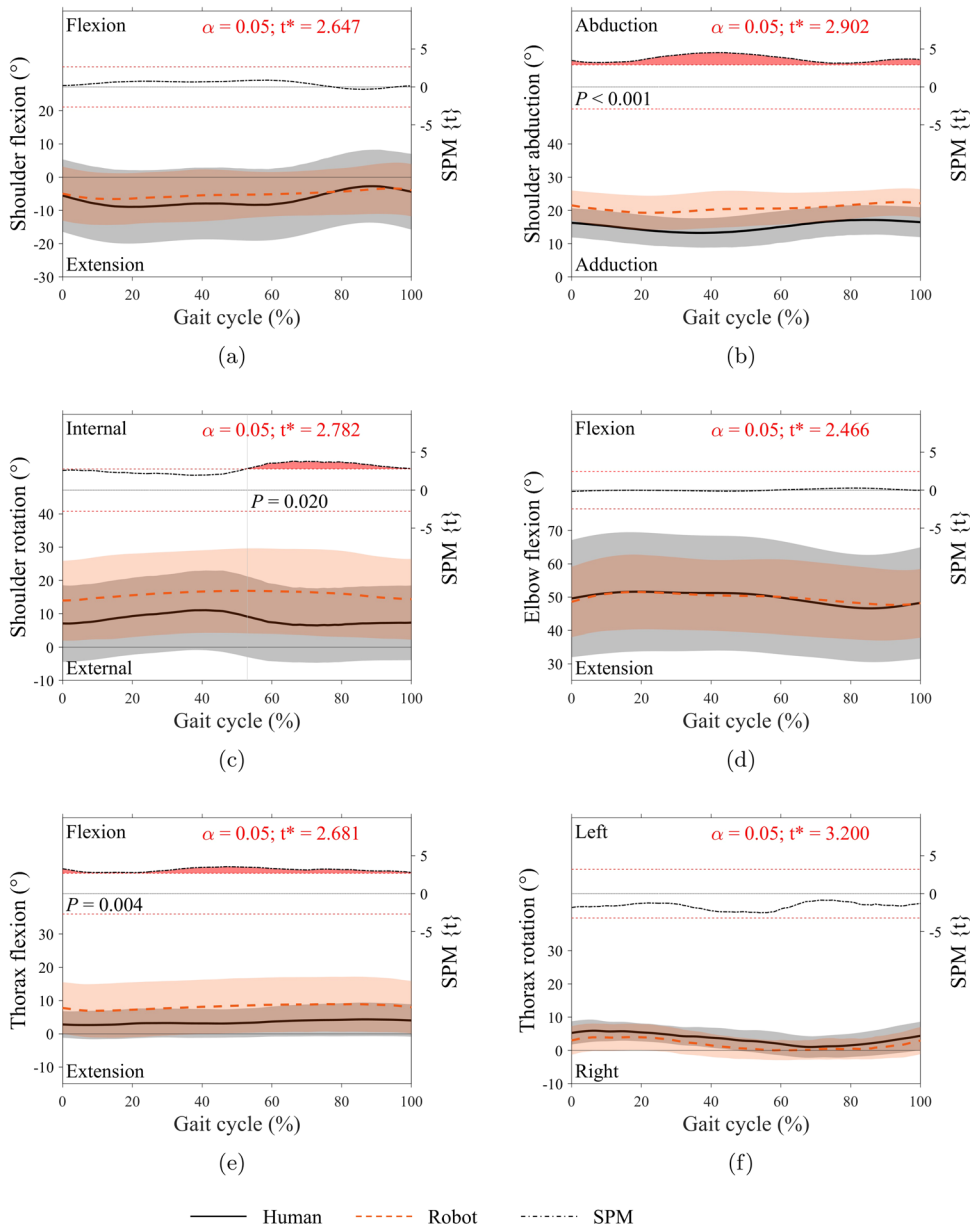


Fig. 3 SPM analysis of upper body. Figures show the mean values of the shoulder flexion/extension (**a**), shoulder abduction/adduction (**b**), shoulder internal/external rotation (**c**), elbow flexion (**d**), thorax flexion/extension (**e**), and thorax left/right rotation (**f**) as a function of normalized gait cycle (left y-axis) with colored area representing the standard deviation in carrying a table with another human (black solid line) or an assistive robot (orange dashed line). In the upper part, the corresponding SPM inference (black dotted line) with the ranges for significantly larger or smaller angles colored red is shown (right y-axis). The statistical significance level (α), critical thresholds (t*) and, in case of significant difference(s), the cluster-level probability value(s) (P) are displayed

27–46 mm for unconstrained walking at similar velocities of 0.6–0.7 m/s (Orendurff et al., 2004; Petrovic et al., 2019).

The described findings for the spatio-temporal parameters, especially shorter steps and slower velocities, further affect the kinematics (Fukuchi et al., 2019; Lim et al., 2017). Given the slow velocity, the slightly flexed hip joint at the beginning of the stance phase of the present study aligns with previously reported results for unconstrained walking (Stansfield et al., 2001; Reznick et al., 2021). Full hip joint extension, as observed during normal, unconstrained walking (Stansfield et al., 2001; Reznick et al., 2021), is not fully achieved in the present study when carrying a table with a human partner, evidenced by a minimum hip joint flexion of 5°. Likewise, the observed knee joint angles of the present study are consistent with previously reported results for normal, unconstrained walking (Stansfield et al., 2001; Reznick et al., 2021). Due to the slow velocity, there is hardly any knee flexion in the first half of the stance phase. With respect to the slow velocity, the ankle joint dorsalextension peak at the end of the stance phase in the present study occurs slightly earlier than for higher velocities, which aligns with reported results for unconstrained walking at different velocities (Stansfield et al., 2001; Reznick et al., 2021). In summary, the findings suggest that when carrying a load collaboratively in front of the body while moving forward, there is a distinct walking pattern compared to walking without constraints. Differences in gait pattern are mainly attributed to the slow walking velocity during collaborative load carrying. Factors like a missing

arm pendulum (Simpkins et al., 2022), the obstructed field of view due to the carried load (Marigold & Patla, 2008), or the slower velocity due to the partner walking backward could have further impacted the gait pattern.

4.2 Human–robot collaborative manipulation

The COM–hand distance showed no significant differences; thus, the table was not carried with more distance in the robotic condition. One could expect a larger COM–hand distance, to increase the distance between the robot and human for safety reasons. For shoulder and elbow flexion, no significant differences between the carrying partners were found (Fig. 3a, Fig. 3d, Table 2). The significantly larger shoulder abduction (SPM: $P < 0.001$; average: $P = 0.002$) with the robotic partner was accompanied by a larger internal shoulder rotation, which differed significantly at the end of the gait cycle ($P = 0.02$). The thorax was significantly more flexed during the entire gait cycle ($P = 0.004$) with the robotic partner, but the average difference was only 2°. As hypothesized, task duration was significantly longer and velocity was significantly slower when carrying the table with the robotic partner ($P < 0.001$, both) (Table 1), which is consistent with the literature (Mielke et al., 2020). When the carrying partner was a robot instead of another human, the spatio-temporal parameters and lower body kinematics were altered more than the upper body kinematics. The literature supports the assumption that the observed differences in the spatio-temporal parameters are mainly the effects of the slower velocity. Due to the slower velocity, step length as well as step frequency were smaller compared to human–human interaction (Table 1) (Fukuchi et al., 2019). Ground contact time was significantly longer ($P < 0.001$) (Table 1). The expected increase of the duty factor for the human–robot interaction was also observed since the velocity was reduced in the human–robot interaction (Fihl & Moeslund, 2007). A significantly larger flexion for the hip joint was found at the end of the stance phase ($P = 0.016$) with the robotic partner, whereas the knee joint was significantly more flexed at the beginning of the stance phase ($P = 0.01$) compared to the human–human interaction (Fig. 2a, c). The ankle joint was significantly more dorsally flexed during the first half of the stance phase ($P = 0.002$) but less at the end of stance phase ($P = 0.007$) (Fig. 2e) compared to the human–human interaction. This leads to the suggestion that the subjects walked in a more flexed posture compared to in the human–human interaction. Additionally, the ROM of the lower limb was significantly smaller ($P = 0.015$, $P < 0.001$, and $P < 0.001$ for the ankle, knee, and hip joints, respectively). The average COM height was not affected by the robot, but the COM amplitude was, on average, 6(6) mm ($P = 0.004$) significantly smaller when carrying the table with the robotic partner. Since walking velocities are known to affect COM oscillation and joint kinematics, the observed differences in the kinematics and COM oscillation might also be effects of the slower velocity. To clearly distinguish whether the causes of the observed differences are the slower velocity during the human–robot interaction, the robot itself or yet unknown reasons, further research is necessary. Furthermore, when interpreting the results, it should be considered that humans interact with a human partner in a variety of ways, e.g., through visual, verbal, and haptic communication, whereas the human–robot interaction in this study was primarily based on haptic information.

It is assumed that maintaining the natural posture of humans and spatio-temporal parameters during a human–human interaction for collaborative human–robot interaction could enhance the comfort and performance of human–robot collaboration. Investigating human posture and spatio-temporal parameters, especially the duty factor and velocity, seems to be a valuable addition to evaluating newly designed robots for collaborative interaction with humans. To facilitate human–like human-robot interaction, the development of future robotic systems designed for interaction with humans may allow for self-selected human velocity, may adapt to human anthropometrics, and may consider the natural oscillation of the human COM.

4.3 Limitations

A few factors deserve further consideration regarding the interpretation of the presented comparison between the human–robot and the human–human collaborative manipulation. As already mentioned, the conditions for the human–human and human–robot interaction were different in terms of walking velocity and the ways of interacting, such as visual, verbal and haptic communication. Further, we do not have any information whether the interaction forces were similar between both interactions and how the load was shared between the carrying partners. Since the ways of communication can affect task duration and walking velocity (Mielke et al., 2020), velocity can affect lower body joint kinematics and spatio-temporal parameters (Fukuchi et al., 2019; Lim et al., 2017), and the interaction forces might affect the kinematics (Hoffman, 2008) it is not possible to clearly distinguish the underlying causes of the observed differences in the kinematics, spatio-temporal parameters, and the oscillation of the COM. Based on the present study, further research should be conducted to understand the effects of velocity, ways of communication between the interaction partners, and interaction forces on the kinematics, spatio-temporal parameters and oscillation of the COM.

Following on from this study, the robot should be tested with the elderly population since it was designed to assist the elderly, who exhibit changes in gait, such as reduced velocity

and step length for comfortable walking velocity (Kerrigan et al., 1998).

4.4 Conclusion

Kinematics and spatio-temporal parameters during a human–human collaborative manipulation task, carrying a load in front of the body moving forward, have been investigated for the first time (to the best of the author's knowledge) in the present study. The results of the human–human interaction can be used as a model for human–robot interaction and provide an important basic knowledge for the development of robots interacting with humans, such as natural oscillation of the COM or the preferred self-selected walking velocity of human dyads. Further research regarding the interaction with the robot is necessary, to fully determine, whether the observed changes of the kinematics and spatio-temporal parameters are solely based on the velocity. While the human-robot interaction was primarily based on haptic information, the robot successfully assisted the human partner in carrying a table.

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Data Availability The data that support the findings of this article are available from the corresponding author upon reasonable request.

Code Availability A custom-madeMATLAB code (MathWorks, Inc., Natick, United States) used for data analysis is available fromthe corresponding author upon reasonable request.

Declarations

Conflict of interest The authors NR, DB, KS, and CW were or are employees of KUKA Deutschland GmbH. The authors declare no other competing financial or non–financial interest.

Ethics Approval The study was approved by the ethics committee of the German Sport University, Cologne (No 132/2020).

Consent to Participate Informed written consent to participate in this study was provided by all subjects.

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